INSTITUTION LOGO

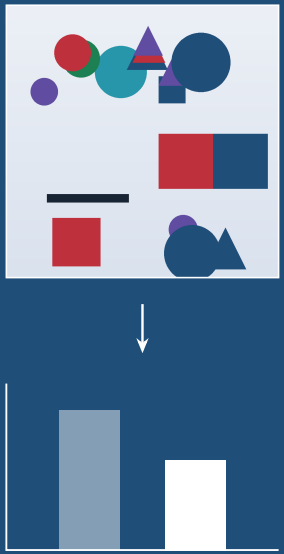
Sharper reconstructions with an edge-aware loss

Author et al., Journal 1, 1 (2026)
https://doi.org/10.1000/182

Your Name

GROUP LOGO

Group name
Department
Institution
Occasion, DD.MM.YYYY



Outline

1	**The problem**	Why does the baseline lose detail?
2	**The method**	How does the method keep edges sharp?
3	**The results**	Does it work?

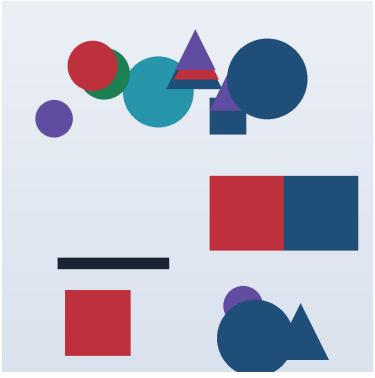
01

The problem

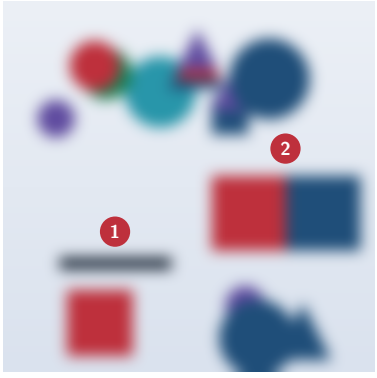
Why does the baseline lose detail?

GROUP LOGO

Where the baseline fails



input



baseline output

1. thin bar fades
2. colours mix along the edge

the loss averages over all pixels; edges are few pixels

GROUP LOGO

Author et al. (2026), Journal 1, 1, Fig. 2a, b

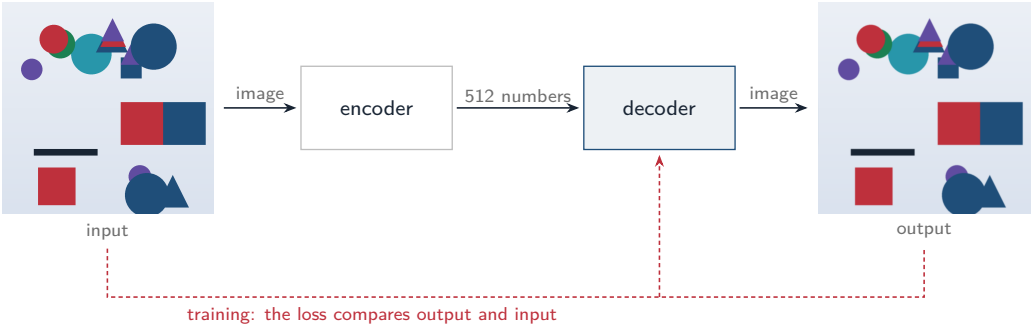
02

The method

How does the method keep edges sharp?

GROUP LOGO

The pipeline



The loss

$$L = L_{\text{data}} + \lambda L_{\text{edge}}$$

per pixel: output against input; the baseline's whole loss

weight of the edge term

image gradients: large along edges

$$\frac{1}{n}\sum_{i=1}^{n}|x_i - \hat{x}_i| \qquad \lambda = 0.1 \qquad \frac{1}{n}\sum_{i=1}^{n}|\nabla x_i - \nabla \hat{x}_i|$$

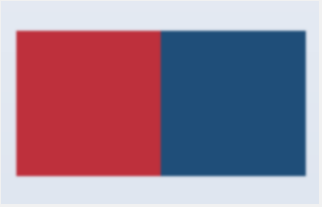
Choosing the weight

$\lambda = 0$, the baseline



- edges blur
- lowest pixel error

$\lambda = 0.1$



- edges sharp
- pixel error close to the baseline's

$\lambda = 1$



- halos along edges
- grain in flat areas

Running it

training and evaluation

```
$ python train.py --config configs/edge.yaml
epoch  1/50   loss 0.412
epoch 50/50   loss 0.087
$ python evaluate.py --split test
error 0.27   baseline 0.42
```

configs/edge.yaml

```yaml
# the loss of slide 4
loss:
  data: l1
  edge: sobel
  weight: 0.1
```

```
project/
  configs/
    edge.yaml      the loss and its weight
  train.py         trains the model
  evaluate.py      the error on a split
```

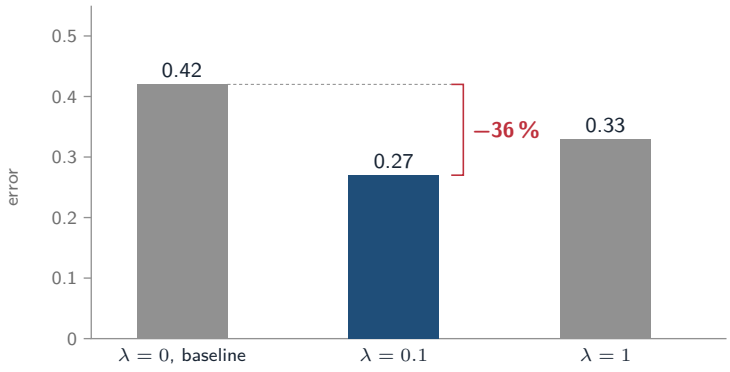
03

The results

Does it work?

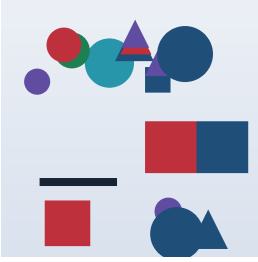
GROUP LOGO

Error on the test set



mean absolute error over the test images; lower is better

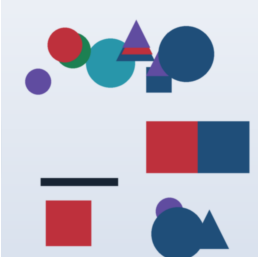
Baseline against method



input



baseline



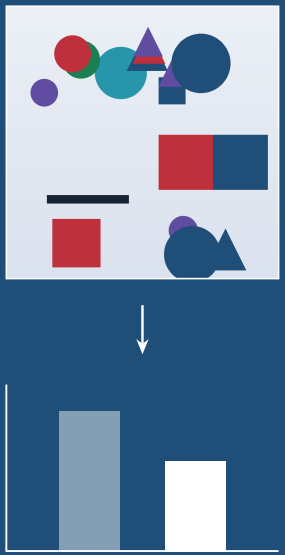
method, $\lambda = 0.1$

- ✓ the thin bar and the shared edge stay sharp
- ⚠ one dataset and one training run so far

INSTITUTION
LOGO

Thank you

Your Name



A1 The loss in full

data term (Eq. 1)

$$L_{\text{data}} = \frac{1}{n} \sum_{i=1}^{n} |x_i - \hat{x}_i|$$

edge term with the Sobel kernels (Eqs. 2, 3)

$$S_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad S_y = S_x^\top$$

$$L_{\text{edge}} = \frac{1}{n} \sum_{i=1}^{n} \left(|S_x * (x - \hat{x})|_i + |S_y * (x - \hat{x})|_i \right)$$

the loss (Eq. 3)

$$L = L_{\text{data}} + \lambda \, L_{\text{edge}}, \qquad \lambda = 0.1$$

on a flat area of both images the gradients vanish, and only L_{data} acts

$$(S_x * x)_i = (S_y * x)_i = 0$$

A2 References

The paper

A. Author, B. Author, C. Author (2026). *Title of the paper*. Journal 1, 1–10. https://doi.org/10.1000/182.

A. Author (2026). *code repository*, commit and licence, read on DD Month YYYY.

Methods

D. Other, A. Author (2020). *Title of the method's paper*. Proceedings of the Conference, 100–110.